

computers were reattached to the network. A few weeks later, a test of the computers using the same virus scanner indicated they had become re-infected with the same viruses. The source of infection was then identified as repositories of executables stored on the Unix file servers.

These repositories were organically grown centralized resources for all the personal computers because the Unix servers were effective at providing these shared services via NFS. In retrospect, this problem had to exist. The use of networked systems that were exported from the Unix platforms provided an easy, powerful method of transferring data, including executables. Some network designs provide all third party software from a network disk for ease of maintenance and reduced storage requirements. This easy access provides an open door for viruses.

Trans-platform Viruses Attack Unix

During late 1994 and early 1995 were observed multiple instances of at least three trans-platform virus attacks on Unix systems. All of these attacks involved MS-DOS viruses that attacked PC based Unix systems. The first attack involved a virus that corrupted the Unix file system every night. The attack was located using a virus scanner and indicated a Unix binary that was executed at midnight by cron. The MS-DOS virus had become embedded in the Unix executable where it was executed. The virus did not perform as designed in that the corruption was the result of the virus attempting to infect other files and was not an intended effect. The virus was reinstalled every morning when the system was restored. The second attack involved an MS-DOS virus that executed and was successful in infecting other files. Once again, the file system corrupted but it took longer in duration, thereby allowing the virus to propagate. The final infection involved a boot sector virus. Since this type of virus executes prior to the loading of the operating system, the differences between Unix and MS-DOS are moot. The PC-BIOS and processor chips are the same in both cases and the virus is able to execute according to design.